

A PROJECT REPORT ON UNIVERSITY PORTAL UI/UX REDESIGN

VIT-AP University Portal as Reference

Course: UI/UX DESIGN LAB

Software Requirements Specification (SRS)

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define and describe the functional and non-functional requirements of the proposed University Portal UI/UX redesign. The system uses the VIT-AP University portal as a functional reference for common academic and administrative services. This document specifies expected system behaviour, constraints, operating conditions and user interactions, and serves as a reference for designers, developers, testers, stakeholders and academic evaluators.

1.2 Scope of this Document

This document covers:

- Functional requirements
- Non-functional requirements
- User characteristics
- System constraints
- Interface requirements
- Performance and security requirements

Detailed implementation and technology-selection decisions are outside the primary scope of this document. The project focuses mainly on the requirements and UI/UX design of the portal.

1.3 Overview

The proposed system is a centralized web-based university portal designed to provide students, faculty and administrators with access to academic and administrative services through a unified interface.

The portal manages and presents information related to:

- Student academic records
- Courses and course registration
- Attendance
- Class timetable
- Marks, grades and examinations
- Fees and payment status
- Leave/outing requests
- University announcements and notifications
- User profiles and support services

The primary goal is to reduce navigation complexity and manual effort by providing a clear, consistent, responsive and accessible digital experience.

1.4 Business / Institutional Context

Universities manage a large number of academic and administrative activities that require frequent interaction between students, faculty members and administrative departments. A university portal acts as the common digital access point for these services. The proposed redesign organizes these services around role-based dashboards and simplified workflows, taking the VIT-AP portal as a reference for the functional domain.

2. GENERAL DESCRIPTION

2.1 Product Functions

- Authenticate users and provide role-based access
- Provide personalized dashboards
- Display course, attendance and timetable information
- Support course registration workflows
- Display marks, grades and examination information
- Provide fee status and supported payment-related access
- Support leave/outing request submission and tracking
- Publish notifications and announcements
- Manage permitted user profile information
- Provide feedback and support facilities

2.2 Similar System Information

The VIT-AP University portal is used as the principal reference system for understanding common university portal services. The proposed design does not aim to reproduce the existing interface exactly. Instead, it focuses on improving information hierarchy, discoverability, consistency, accessibility, responsiveness and the efficiency of frequent user tasks.

2.3 User Characteristics

User Type	Characteristics
Students	Basic computer/mobile and internet literacy; frequently access academic and administrative services.
Faculty	Familiar with digital academic systems; manage course, attendance and student-related information.
Administrators	Authorized personnel with knowledge of institutional processes and portal management.

2.4 Problem Statement

University portals may become difficult to use when services are distributed across many menus, screens and workflows. Common usability problems can include:

- Complex navigation and difficulty locating frequently used services
- Information-heavy screens with weak visual hierarchy
- Inconsistent interaction patterns between modules
- Extra steps for routine academic tasks
- Limited usability on smaller screens

The proposed UI/UX redesign addresses these issues by centralizing important information, simplifying navigation and creating consistent workflows for common university activities.

2.5 User Objectives

- Students want quick access to attendance, timetable, courses, marks, examinations, fees and requests.
- Faculty want efficient access to assigned courses, schedules and permitted academic management functions.
- Administrators want structured management of users, academic information, announcements and workflows.
- All users require a clear, reliable and responsive portal with minimal unnecessary navigation.

2.6 General Constraints

- Web-based access is required for the proposed portal.
- Secure authentication is mandatory for protected functions.
- Access to modules is controlled according to user role and permissions.
- Academic information depends on accurate institutional data.
- The portal must support standard desktop and mobile browsers.
- Sensitive academic and personal information must be handled securely.

3. FUNCTIONAL REQUIREMENTS

FR1: User Authentication

Description: The system shall allow authorized students, faculty members and administrators to log in using valid university credentials. The system shall direct authenticated users to the appropriate role-based dashboard.

Priority: High **Dependencies:** None

FR2: Role-Based Dashboard

Description: The system shall display a personalized dashboard containing important academic or administrative information, shortcuts, alerts and recent updates relevant to the authenticated user's role.

Priority: High **Dependencies:** FR1

FR3: Course and Registration Management

Description: The system shall allow students to view available/registered courses and access permitted course registration functions. Faculty and administrators shall access course information according to their permissions.

Priority: High **Dependencies:** FR1

FR4: Attendance Management

Description: The system shall allow students to view course-wise attendance information and shall allow authorized faculty to manage attendance records where applicable.

Priority: High **Dependencies:** FR1, FR3

FR5: Timetable Management

Description: The system shall display class schedules with available course, date/time and venue information. Users shall be able to access their relevant timetable from the portal.

Priority: High **Dependencies:** FR1, FR3

FR6: Marks and Grades

Description: The system shall allow students to view published assessment marks, grades and result information for applicable courses and semesters.

Priority: High **Dependencies:** FR1, FR3

FR7: Examination Management

Description: The system shall provide examination-related information such as schedules, course details, venue information and available examination notices.

Priority: High **Dependencies:** FR1, FR3

FR8: Fees and Payment Information

Description: The system shall allow students to view applicable fee information, payment status and available transaction/receipt information, and provide access to supported payment actions where applicable.

Priority: High **Dependencies:** FR1

FR9: Leave / Outing Request Management

Description: The system shall allow eligible students to submit leave or outing requests with required details and track the approval status of submitted requests.

Priority: Medium **Dependencies:** FR1

FR10: Notifications and Announcements

Description: The system shall display academic and administrative announcements and notifications. Important or time-sensitive information shall be visually distinguishable.

Priority: High **Dependencies:** FR1

FR11: Profile Management

Description: The system shall allow users to view their profile and update only those fields that are permitted by university policy.

Priority: Medium **Dependencies:** FR1

FR12: Feedback and Support

Description: The system shall allow users to submit feedback or support requests and receive a submission acknowledgement. Request status may be displayed where supported.

Priority: Medium **Dependencies:** FR1

FR13: Logout and Session Termination

Description: The system shall allow authenticated users to securely log out and terminate the active portal session.

Priority: High **Dependencies:** FR1

4. INTERFACE REQUIREMENTS

4.1 User Interface

- Web-based graphical user interface with role-based dashboards
- Consistent sidebar/top navigation for major portal modules
- Cards or summary components for important dashboard information
- Readable tables and forms for academic data
- Clear online help, validation, success, warning and error messages
- Responsive layouts for desktop, tablet and smartphone screens
- Visible loading, empty and system-status states

4.2 Hardware Interfaces

- Desktop computer
- Laptop
- Tablet
- Smartphone
- No specialized hardware required for normal portal access

4.3 Communication Interfaces

- Secure HTTPS-based communication
- University network/internet connectivity
- Email or portal notification services where applicable
- Secure integration with approved payment services where applicable

4.4 Software Interfaces

- University authentication service
- Academic records/course information systems
- Attendance and examination data services
- Fee/payment services where applicable
- Notification/email services
- Institutional database/backend services

5. PERFORMANCE REQUIREMENTS

- Frequently used portal pages should load within approximately 3 seconds under normal operating conditions.
- The portal should support a large number of concurrent university users during peak periods such as registration or result publication.
- User actions such as request submission should provide immediate acknowledgement or a visible processing state.
- The system should target high operational availability except during planned maintenance or unavoidable outages.
- The architecture and interface should remain usable as the number of users, courses and records increases.
- Updated statuses such as approvals or payment confirmations should be reflected promptly when the supporting service provides the data.

6. OTHER NON-FUNCTIONAL REQUIREMENTS

6.1 Security

- Secure authentication and session management
- Role-based access control
- Protection of personal and academic information
- Secure transmission of sensitive information

6.2 Reliability

- Regular backup of critical institutional data in a production system
- Graceful handling of service failures

- Prevention of accidental loss or corruption of user actions

6.3 Maintainability

- Modular portal structure
- Reusable and consistent UI components
- Easy updates and bug fixes
- Clear documentation

6.4 Portability

- Accessible through standard modern browsers
- Platform-independent web interface

6.5 Extensibility

- Support for future modules and university services
- Design adaptable to future mobile applications or new integrations

6.6 Reusability

- Common components such as navigation, forms, cards, tables and alerts should be reusable across modules

6.7 Customer / User Support

- FAQs or help information
- Feedback/support forms
- Support request acknowledgement and tracking where available

6.8 Logging and Audit Trail

- Secure logging of important authentication and administrative actions
- Auditability of significant record changes where required

6.9 Application Compatibility

- Compatible with common desktop and mobile browsers
- Responsive across common screen sizes

6.10 Resource Utilization

- Efficient use of network bandwidth and server resources
- Avoid unnecessary loading of large interface assets

6.11 Serviceability

- Authorized administrators should be able to maintain and troubleshoot portal services
- Clear error information should support issue resolution

6.12 Usability and Accessibility

- Consistent navigation and terminology
- Clear visual hierarchy and readable typography
- Keyboard-accessible interactions where applicable
- Sufficient contrast and visible focus/error states
- Minimal steps for frequent student and faculty tasks

7. UI/UX DESIGN REQUIREMENTS

- The dashboard shall prioritize frequently used and time-sensitive information.
- Primary navigation shall group related academic and administrative services logically.
- The current page or selected module shall be visually identifiable.
- Forms shall use clear labels, required-field indicators and contextual validation.
- Important actions shall provide confirmation, success or error feedback.
- Tables shall be readable, scannable and responsive where practical.
- The design shall maintain consistent spacing, typography, icons, buttons and component behaviour.
- The interface shall avoid unnecessary visual clutter and reduce cognitive load.
- Mobile layouts shall preserve access to essential portal functions.
- The redesign shall focus on improving the user experience rather than exactly copying the existing VIT-AP portal UI.

8. MODULE DELIVERABLES

8.1 Student Module

- Authentication and dashboard
- Courses and course registration
- Attendance
- Timetable
- Marks and grades
- Examinations
- Fees/payment information
- Leave/outing requests
- Notifications and announcements
- Profile
- Feedback/support
- Logout

8.2 Faculty Module

- Authentication and faculty dashboard
- Assigned courses and timetable
- Attendance management where authorized
- Permitted academic/student information
- Announcements
- Profile
- Logout

8.3 Administrator Module

- Administrator authentication and dashboard
- User and role management
- Academic/course information management
- Announcements and portal information
- Administrative workflow support
- Feedback/support management

- Portal configuration and audit access
- Logout

9. CONCLUSION

The proposed University Portal UI/UX redesign provides a requirements foundation for a centralized, role-based academic and administrative portal. By using the VIT-AP University portal as a functional reference and applying clearer navigation, consistent components, responsive design and accessibility principles, the project aims to make common university tasks easier to discover and complete. These requirements can be used as the basis for information architecture, user flows, wireframes, prototypes and usability evaluation.